

Decoding Kryptos K1 Using a “no-math(s)” logical method

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Analyzing the Kryptos tableau panels to determine cipher

In a standard Vigenère tableau (tabula recta), the letters begin A-Z for the first encryption/decryption line, and is shifted by 25 characters forward (1 character back) for each subsequent line. The plaintext is then encoded by a repeating keyword that serves as the lookup table. The decode is simply utilizing the same key in the opposite direction to obtain the plaintext. For the Kryptos tableau panels, it begins with the A-Z order on the first line, but the second line the letters KRYPTOSABCDE are used. This modification to the alphabetic order is utilized for Quagmire family ciphers.

Quagmire Ciphers and KRYPOS in tableau panels

Unlike the standard Vigenère cipher, the Quagmire family uses dual key system of a ciphertext key (KRYPTOS - horizontal) and an indicator key (i.e. PALIMPSEST - vertical).

Attached in appendices are the detailed breakdown of all four Quagmire variants. This can be summarized below:

	Plain alphabet keyword	Cipher alphabet keyword	Indicator key (column placement)
Quagmire I	✓	N/A	Intersects at last letter of plain alphabet keyword
Quagmire II	N/A	✓	Key at column 0
Quagmire III	✓	✓	Key at column 0
Quagmire IV	✓	✓ (different key)	Key at column 0

The tableau on Kryptos shows the lettering of ABCDEF...BCD on the top row which may suggest Quagmire II. Since there is a cipher alphabet keyword of KRYPTOS, this would eliminate Quagmire I (no cipher alphabet keyword). Quagmire IV can also be eliminated because the cipher alphabet keyword is different from the plain alphabet keyword. For now, the cipher could Quagmire II or III.



Figure 1 – WIRED 2005 interview diagram
<https://www.wired.com/2009/04/ff-kryptos/>

Determining key length from repeating segments

What is important to note for the Quagmire and Vigenère ciphers is that there is a repetition of the key throughout the ciphertext. That is, eventually certain groups such as THE or OF will begin to repeat itself as the ciphertext length increases. By measuring the distance between the repeating segments, one can determine a possible key length. With the observation VJYQT repeating from position 31-35 and 51-55, there appears to be a period of 5, 10 or 20 to the key. This means the key itself is likely a word of 5, 10, or 20 characters in length¹.

Stenographic clues

While it is possible to try all three key possibilities, a stenographic clue in the tableau greatly aids in the process to determine a key length of 10 characters. By examining the tableau in greater detail, the P on the 2nd line and G on the 12th line of the tableau is sticking out, giving hint to the size of the tableau (11 characters from PTOSABCDEFGH on the right side). What is most critical about stenographic clues is that the information from the artist must be retained in all iterations of CIA Kryptos and mini-Kryptos versions (nickel-silver & copper editions). The WIRED 2005 version also may also originate from the artist as the blueprint for mini-Kryptos (see above). The original CIA Kryptos has the P & G of the tableau sticking out on the right, and the small copper Kryptos in Jim Sanborn's studio during the Great Big Story (2019) video has the P & G leftwards. Regardless of the positioning, the P & G provide vital

stenographic information on the size of the Quagmire tableau.

Combining both the tableau analysis, cyclic periodicity, and stenographic clue (11 characters) would indicate that Quagmire III is used and row 2 is the beginning of the plaintext alphabet keyword. Quagmire III will use KRYPTOSABC in contrast to Quagmire II ABCDEFG (no plain alphabet keyword).



Figure 2 – Jim Gillogly photo, Jim Sanborn studio Kryptos model and nickel-silver mini Kryptos all with the stenographic P:G clue on the top left tableau giving rise to the Quagmire III tableau.

¹ Assuming the indicator key is an actual English word

Quagmire III Attack

Mapping Out the Ciphertext by Characters and Rows

With the possible cipher period of 10 determined, the next step is to map out all the characters for the ciphertext relative to the decryption tableau. When it comes to analog encryption systems, there is a trial and error component that will be required.

1	2	3	4	5	6	7	8	9	10	
E	M	U	F	P	H	Z	L	R	F	
11	12	13	14	15	16	17	18	19	20	
A	X	Y	U	S	D	J	K	Z	L	
21	22	23	24	25	26	27	28	29	30	
D	K	R	N	S	H	G	N	F	I	
31	32	33	34	35	36	37	38	39	40	
V	J	Y	Q	T	Q	U	X	Q	B	
41	42	43	44	45	46	47	48	49	50	
Q	V	Y	U	V	L	L	T	R	E	
51	52	53	54	55	56	57	58	59	60	
V	J	Y	Q	T	M	K	Y	R	D	
61	62	63								
M	F	D								

		0	1	2	3	4	5	6	7	8	9	0	1	1	1	1	1	1	1	2	2	2	2	2	0	1	2	3			
		A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
0	A	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P
1	B	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T
2	C	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O
3	D	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S
4	E	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A
5	F	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B
6	G	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C
7	H	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D
8	I	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E
9	J	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F
10	K	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G

Each ciphertext character is mapped out to the tableau and color coded to illustrate the banding patterns. The ordering of the cipher alphabet lines is in the same order as the stenographic tableau for ease of reference. This is not the final orientation of the cipher lines as each line must be translated so that the indicator key will line up at column 0. At this moment, the indicator key is not known but there are elements of the English language that can be utilized to greatly reduce the possibilities.

Principal of Symmetry

One of the weaknesses of symmetric encryption is that the same key is used for encoding and decoding. With the advent of computers, the key itself can be brute forced through testing of all possibilities. In a manual pencil and paper method, the symmetrical component is the relationship of the indicator line ciphers to the plaintext as well as the cipher line position relative to other letters. In the case of K1, the Quagmire III cipher has been identified so that means the order of the plaintext line will be

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z

As shown above there is a 6-character shift from K to S. Moreover, the ordering is maintained through the entire plaintext alphabet and ciphertext alphabet. Unfortunately, no plaintext crib is available to aid in the deduction of the cipher alphabet lines 1 through 10.

Vowel Distances to Deduce Cipher Alphabet Line Shifts

With the alphabet of 26 characters, the distances between letters can be classified as short (1-7 shifts) medium (8-13 shifts) and long (14-25 shifts). For example, A to Z on the alphabet line is from 25 to 7 or 18 shifts. While it may seem useful to identify long chains, the issue is that the inverse holds true. A short chain in the opposite direction may yield a different result. That being said, the medium distances will remain medium length for its inverse direction.

While there are many possible relationships that would produce medium chains, the focus was on the usage of vowels on the cipher alphabet lines.

					0	1	2	3	4	5	6	7	8	9	10										
K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z

The relationship of O:A is 2 shifts, A:E is 4 shifts and E:I being 4 shifts. Combining the shifts medium chains can be produced from A:I of 8 shifts to the right (10-2) and O:I of 10 shifts to the right (10-0). When this is mapped out to the cipher alphabet lines, we get the following possibilities.

LINE	A:I (8 SHIFTS)	O:I (10 SHIFTS)	A:I & O:I SHIFT POSSIBLE
1B	D:M 8 shifts		
2C	J:X 8 shifts & M:K 8 shifts	F:V 10 shifts & J:K 10 shifts	J:X:K possible (A:I & O:I) F:V 10 shift possible (O:I only) J:X 8 shift (A:I only)
3D			
4E	F:Q 8 shifts		
5F	V:T 8 shifts		
6G	D:M 8 shifts	D:Q 10 shifts	D:M:Q possible (A:I and O:I)
7H	G:U 8 shifts & L:Z 8 shifts		
8I			
9J	F:Q 8 shifts		
10K			

Now the trial and error portion begins by assuming all 8 shifts are A:I and all 10 shifts fall under O:I the cipher alphabet lines will produce the following plaintext. For 2C and 6G, both O:I 10 shifts also have to hold true.

1	2	3	4	5	6	7	8	9	10	
B	A/E/C		A	E	E	N/G		Q		
11	12	13	14	15	16	17	18	19	20	
I	G/L/I		J	H	A	D/T		M		
21	22	23	24	25	26	27	28	29	30	
A	I/N/L		H	H	E	A/R		O		
31	32	33	34	35	36	37	38	39	40	
N	O/C/A		I	F	L	I/C		G		
41	42	43	44	45	46	47	48	49	50	
L	E/I/G		J	T	H	E/S		Q		
51	52	53	54	55	56	57	58	59	60	
N	O/C/A		I	F	I	Q/H		Q		
61	62	63								
I	R/O/P									
P	E/A/C		O	M	P	S/F		A		

Repeating segments using THE and OF

When repeating segments are identified, it is possible that some of the characters used may represent “the” or “of” which is commonly used in the English language. In the case of K1, it is identified that there is a repeating segment of VJYQT (31-35 & 51-55) which might possibly utilize a section of OF or THE.

Looking at the cipher alphabet lines, position 45-47 is T-H-E/S. There is a possibility that position 47 might be the E in THE. By using the English language and intuition, positions 4-7 indicate E-E-N/G. There are only a few words that use EEG such as *squeegie* or *geegaw* compared to EEN. Likely position 7 of the cipher lines use S rather than F as the indicator line. Furthermore, examining positions 24-26 there is a HHE and 34-35 IF as well as 54-55 IF this may also give clues to a THE and some OF words. It is likely the indicator line 4 of O is incorrect because of the AEE in positions 4-6. By adjusting the indicator lines 7 (use S) and line 4 to produce THE & OF, N24 must map to plaintext T and Q34 and Q54 must map to O. Indicator line 4 can be manipulated to I on column 0 to produce.

1	2	3	4	5	6	7	8	9	10	
B	A/E/C		W	E	E	N		Q		
11	12	13	14	15	16	17	18	19	20	
I	G/L/I		S	H	A	D		M		
21	22	23	24	25	26	27	28	29	30	
A	I/N/L		T	H	E	A		O		
31	32	33	34	35	36	37	38	39	40	
N	O/C/A		O	F	L	I		G		
41	42	43	44	45	46	47	48	49	50	
L	E/I/G		S	T	H	E		Q		
51	52	53	54	55	56	57	58	59	60	
N	O/C/A		O	F	I	Q		Q		
61	62	63								
I	R/O/P									
P	E/A/C		I	M	P	S		A		

The letters WEEN appear from positions 4-7 after changing indicator line 4 to I. There are not many words that use WEEN such as *sweeney*, *tween*, and if position 1 is incorporated, the word BETWEEN is possible if indicator line 2 uses the letter A. Using the word BETWEEN to adjust indicator lines 2 & 3:

1	2	3	4	5	6	7	8	9	10	
B	E	T	W	E	E	N		Q		
11	12	13	14	15	16	17	18	19	20	
T	L	E	S	H	A	D		M		
21	22	23	24	25	26	27	28	29	30	
A	N	D	T	H	E	A		O		
31	32	33	34	35	36	37	38	39	40	
N	C	E	O	F	L	I		G		
41	42	43	44	45	46	47	48	49	50	
L	I	E	S	T	H	E		Q		
51	52	53	54	55	56	57	58	59	60	
N	C	E	O	F	I	Q		Q		
61	62	63								
I	O	N								
P	A	L	I	M	P	S		A		

Once the indicator key of PALIMPS*(A)* appear, it is likely an English word that can be searched through the dictionary with PALIMPS. The only words available are PALIMPSEST and PALIMPSESTS. Conveniently indicator lines 8-10 can be filled with the 10-letter word EST portion of PALIMPSEST to obtain the final K1 decode.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y
A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S
L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J
I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H
M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L
P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y
S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O
E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D
S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O
T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P

1	2	3	4	5	6	7	8	9	10	
B	E	T	W	E	E	N	S	U	B	
11	12	13	14	15	16	17	18	19	20	
T	L	E	S	H	A	D	I	N	G	
21	22	23	24	25	26	27	28	29	30	
A	N	D	T	H	E	A	B	S	E	
31	32	33	34	35	36	37	38	39	40	
N	C	E	O	F	L	I	G	H	T	
41	42	43	44	45	46	47	48	49	50	
L	I	E	S	T	H	E	N	U	A	
51	52	53	54	55	56	57	58	59	60	
N	C	E	O	F	I	Q	L	U	S	
61	62	63								
I	O	N								
P	A	L	I	M	P	S	E	S	T	

Conclusion

K1 can most definitely be solved through a pencil and paper method, but will require grid paper, scissors (for Caesar shifts of each indicator line) and a dictionary to look up the word palimpsest. Codebreakers are extremely fortunate that Jim Sanborn used the English language for the Quagmire III keys and without it, likely the brute-force autokey method would be necessary. Some of the weaknesses exploited on the Quagmire III solution were:

1. Cyclical nature of encryption (repeated use of the key) to results in repeated segments – the longer the ciphertext the most likely the words OF and THE will appear
2. Symmetrical nature of polyalphabetic substitution – for each indicator line, the ciphertext and plaintext is both decoded and encoded by the same key
3. Relationships of alphabet characters to each other (shifts between characters are preserved in plaintext and ciphertext alphabet ordering)
4. English words used as keys to the cipher – the advent of computing can brute force such ciphers

The bottom half of the tableau might also be useful for the decryption of K2 as the top half was for K1.

K4 Prediction

K4 is likely a multi step encryption that may use a substitution cipher of some sorts. To increase the difficulty, the weaknesses identified above can be mitigated by:

1. Non-repeating key at least to the length of the plaintext
2. An asymmetrical encryption method where the encoding key is different from the decoding key
3. Independent positional character shifts (not in relation to any other text positions)
4. Non-English key akin to gibberish AND
5. Extra concealment or masking of the plaintext as phonetically spelled words (four as P-H-O-A-R)

All of these weaknesses can be mitigated with a one-time pad, XOR, or Vernam cipher. To increase the difficulty, there would likely be an unconventional K4 transposition required prior to the decode. The best guess is that K4 is literally a public key OTP decode key that must be manipulated through transposition (and or substitution) to decrypt K4. This would be akin to a combination lock or a word lock where the key is merely scrambled.

The BERLIN and CLOCK clue from 2010 and 2014 respectively provides hints that if a one-time pad decode for NYPVTT and MZFPK is required for positions 64-69 and 70-74 respectively, the decode key would simply be MUYKLGKORNA. Coincidentally there is only a single M and Y in K4. If the K1 intuitive method shown above was any indication to the approach of K4, it is that some form of shifting or rotating might be incorporated on the decoding to significantly reduce the need for complex mathematics. The most elegant transpositional method would be utilizing a Jefferson disk or M-94 military cipher to manipulate the K4 public key prior to OTP decoding.

Quagmire I - Plain alphabet keyword (KRYPTOS) and indicator key (CIPHER) intersection

[illegible][illegible]

K	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	O	Q	R	S	T	U	V	W	X	Y	Z
W	C	J	X	D	A	F	G	M	N	E	K	R	J	I	P	O	W	L	M	S	T	A	S	P	C	D	Q	R	E	F
J	B	C	K	C	Z	M	N	E	K	R	J	I	P	O	W	L	M	S	T	A	S	P	C	D	Q	R	E	F	G	H
Y	B	B	H	O	G	D	Q	L	F	G	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L

A	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
K	R	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
K	R	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
K	R	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
K	R	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
K	R	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z

[illegible]

A	B C D E F G H I J K L M N O P Q R S T U V W X Y Z																								
	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
CIPHER	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
	J	T	O	S	L	M	I	A	B	C	Q	D	U	V	N	X	Z	E	F	G	H	I	J	K	L
	I	T	O	S	L	M	I	A	B	C	Q	D	U	V	N	X	Z	E	F	G	H	I	J	K	L
	Y	P	T	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B

K	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z

[illegible]

K	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
C	D	E	F	G	H	I	J	L	M	N	X	Q	U	V	W	X	Z	K	R	A	B	C	D	E	F
I	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	R	A	B	C	D	E	F	G
P	J	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	R	A	B	C	D	E	F	G
H	I	F	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X
E	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z

K	R	Y	P	T	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30		
K	1	2	3	4	5	6	7	8	9	B	C	D	E	F	G	H	I	J	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
K	E	E	Y	Y	W	W	R	D	A	B	C	F	G	H	I	J	L	M	N	P	Q	R	S	T	U	V	W	X	Y	Z	Z	Z	Z
K	E	E	Y	Y	W	W	R	D	D	B	B	C	F	G	H	I	J	L	M	N	P	Q	R	S	T	U	V	W	X	Y	Z	Z	Z
K	E	E	Y	Y	W	W	R	D	D	B	B	C	F	G	H	I	J	L	M	N	P	Q	R	S	T	U	V	W	X	Y	Z	Z	Z
K	E	E	Y	Y	W	W	R	D	D	B	B	C	F	G	H	I	J	L	M	N	P	Q	R	S	T	U	V	W	X	Y	Z	Z	Z
K	E	E	Y	Y	W	W	R	D	D	B	B	C	F	G	H	I	J	L	M	N	P	Q	R	S	T	U	V	W	X	Y	Z	Z	Z

[illegible]

K	1 2 3 4 5					6 7 8 9 10					11 12 13 14 15					16 17 18 19 20					21 22 23 24 25					
	R	P	T	O	S	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
CIPHER	F	G	H	I	J	K	L	M	N	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
I	J	K	L	M	N	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
Y	W	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

Quagmire III - Decoding Tableau with Shifts

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25																									
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	
C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z		
D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z			
E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z				
F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z					
G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z						
H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z							
I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z								
J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z									